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[2510] 2025/2026 Semester 1

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S1AY2526 Assignment 2

>>> Welcome to the Sotong Game <<<

Image generated by DALL·E through ChatGPT, OpenAI. Prompt authored by Ni Qingping. Generated on October 12, 2025.]

Background

When the sun slips behind the HDB blocks and the kopitiam shutters rattle, the old estate turns into a playground of whispers and neon. Chalk lines bloom under corridor lights, and a tiny "referee" wakes on each block. No microcontroller, steady as a heartbeat. Two rules step into the ring: the **Players**, light on their feet. The **Enforcers**, eyes sharp as cats. The night isn't about speed alone, it's about composure and good judgment. Stay under the lights freeze, and who can read the alleyways when the chase begins.

Game 1: Red Light, Green Light:

The estate holds its breath. Fans hum, lifts ding, and a single blink decides fate: **dash on green, become a statue on red**. The best runners aren't the fastest, but the calmest. The ones who can silence a tremor when the whole block is watching. Your board will sense the slightest movement and call it fairly.

Game 2: Catch & Run:

The chalk lines turn to back alleys and void decks. **Players weave, hint, and double back. Enforcers close in and strike.** There's no final flag tonight, only survival. Outlast the count, slip past the sweep, don't get tagged. Your board will feel the night, and keep the pace with clean, visible signals so the games can go on till the streetlights yawn.

Two games, two modes. And one tiny brain, give it senses. Keep calm, keep running, let the Sotong game begin :)

Overview

Your board represents the **Referee Unit**, controlling and monitoring two games:

- **Game 1 – Red Light, Green Light**
- **Game 2 – Catch & Run**

Each game tests different aspects of awareness, precision, and stability. Players are tracked using onboard sensors, and outcomes are indicated through LED patterns and UART messages. The user-push button (PB0) controls mode transitions. Each group can choose to be a Player or an Enforcer.

Peripherals Used

- **User Push button (PB):** Used to switch modes or confirm game events.
- **Temperature, Humidity and Pressure Sensor:** Simulate environmental changes in the surroundings (e.g., humidity spike = tension or chase intensity). Used as secondary thresholds.
- **Accelerometer & Gyroscope:** Detects player movement or tilt: critical in Red Light, Green Light for judging if the player moved during "Red Light."
- **Magnetometer:** Acts as a proximity detector to the Enforcer during Catch & Run. Stronger readings mean the Enforcer is closer.
- **UART Transmission**: Sends updates, warnings, and status to the referee unit.
- **LED**: Visual game indicator based on phase and outcome.

Modes of Operation

Game 1 – Red Light, Green Light

When powered on, the device enters **Red Light, Green Light** as the default starting mode.

1. **Startup Message via UART:**
   - `"Entering Red Light, Green Light as Player/Enforcer"`
2. **Game Overview:**
   - Every 10 seconds, the game command alternates between "**Green**" and "**Red**" light phases (not the LED2 colour).
   - During **Green Light**, players can "move" freely.
   - During **Red Light**, the accelerometer and gyroscope monitor for any movement above a defined threshold of your choice.
3. **Gameplay (Green Light):**
   - **For both PLAYER and ENFORCER:**
     - UART
       - Sends once: `"Green Light!"`
       - Environment monitoring: Temperature, Pressure and Humidity sensors readings transmitted via UART every 2 seconds.
     - LED: Always ON.
4. **Gameplay (Red Light):**
   - **For both PLAYER and ENFORCER:**
     - UART:
       - Sends once: `"Red Light!"`
       - Movement monitoring: Accelerometer and Gyroscope readings transmitted via UART every 2 seconds.
     - LED: Toggles every 0.5 seconds.
   - **PLAYER**: If the Accelerometer and Gyroscope readings exceed thresholds of your choice
     - UART sends once: `"Game Over!"`
     - Game ends
   - **ENFORCER**: If the Accelerometer and Gyroscope readings exceed thresholds of your choice
     - UART sends once: `"Player Out!"`
     - Game continues
5. **Switch Game:**
   - **Double-press user pushbutton**: Switch to *Catch & Run (Game 2).*

Game 2 – Catch & Run

Once switched, the unit enters **Catch & Run Mode**, a chase simulation between Player and Enforcer.

1. **Startup Message via UART:**
   - `"Entering Catch And Run as Player/Enforcer"`
2. **Game Overview:**
   - Magnetometer continuously checks the magnetic field strength, signifying the proximity between the player and the enforcer.
   - The player is trying to run from the enforcer, and the enforcer is trying to catch the player
   - Environment Monitoring (for both Player and Enforcer): Temperature, Pressure and Humidity sensors are read every 1 second but **only transmitted via UART when the readings exceed defined thresholds**. Transmission stops when the values return to normal. Example:
     - `"Temperature spike detected! 7.15C. Dangerous environment!"`
3. **Gameplay (PLAYER):**
   - When the magnetic field exceeds a defined threshold (simulating the Enforcer nearing the Player):
     - UART: `"Enforcer nearby! Be careful."`
     - LED blinks faster as proximity increases. (You may implement your own blinking effect. A minimum of 3 different blinking frequencies should be used).
   - If user pushbutton single-press is **detected within 3 seconds** of threshold violation of magnetometer
     - UART: `"Player escaped, good job!"`
     - Game continues
   - If user pushbutton single-press is **NOT detected within 3 seconds** of threshold violation of magnetometer
     - UART: `"Game Over!"`
     - Game ends
4. **Gameplay (ENFORCER):**
   - When the magnetic field exceeds a defined threshold (simulating the Enforcer nearing the Player):
     - UART: `"Player is Nearby! Move faster."`
     - LED blinks faster as proximity increases. (You may implement your own blinking effect. A minimum of 3 different blinking frequencies should be used).
   - If user pushbutton single-press is **detected within 3 seconds** of threshold violation of magnetometer
     - UART: `"Player captured, good job!"`
     - Game continues
   - If user pushbutton single-press is **NOT-detected within 3 seconds** of threshold violation of magnetometer
     - UART: `"Player escaped! Keep trying."`
     - Game continues
5. **Switch Game:**
   - **Double-press user pushbutton**: Switch to *Red Light, Green Light (Game 1)*

General Clarifications and Hints

- For a project of this nature, it is impossible to have the specifications cover all possible scenarios. We leave it to your discretion to decide how the system should respond in scenarios not covered by the above specifications.
- The exact messages/format displayed on the PC is not important. The format and messages are just suggestions. Feel free to make reasonable modifications.
- While you are free to set the various thresholds based on experimentation (i.e., empirically), you should be able to explain them in terms of the physical quantities or units involved.

Assignment Submission

The Assignment 2 Canvas submission deadline is as shown below. The assessment schedule will be uploaded to Canvas. **No late submission is considered, try NOT to do last-minute submissions to prevent traffic congestion.** You must submit your exported project folder as a .zip file (Archive instructions given below) and submit your slides as a .ppt file.

Lab Class	Deadline for Canvas submissions
EE2028-B01 (Tuesday AM Session)	9:00 AM, 4-NOV
EE2028-B04 (Tuesday PM Session)	2:00 PM, 4-NOV
EE2028-B05 (Wednesday PM Session)	2:00 PM, 5-NOV
EE2028-B02 (Thursday AM Session)	9:00 AM, 6-NOV
EE2028-B03 (Friday PM Session)	2:00 PM, 7-NOV

Please follow the rules here to name your archived project and slides (ONE set of submission per group):

Get your group number from Canvas, and name the project file and slides that are to be submitted (DO NOT zip them together) as the following naming templates:

- **For the Project (only .zip file is accepted):**

Assignment1_Group Number_ Matriculation number of all team members separated by an underscore_Code.zip

Example: Assignment2_12_A0213101Z_A0010101Z_Code.zip

- **For the Slides (only .ppt file is accepted):**

Assignment1_Group Number_ Matriculation number of all team members separated by an underscore_Slides.ppt

Example: Assignment2_12_A0213101Z_A0010101Z_Slides.ppt

The slides have no page limit, and you don't need to attach the entire code as an appendix. Please include the following in your slides and presentation:

- Flowchart(s) showing your system design;
- Detailed implementation of your code, including your overall logic and functions;
- Enhancement/improvement you have done in addition to the compulsory requirements;
- Problems you have encountered during this assignment and how did you manage to solve them;
- (Optional) An Appendix for any feedback or suggestions you would like to give to the teaching team.
- and an Appendix that declares every member's joint and specific individual contributions towards this assignment.

Please prepare a 15-minute presentation for the assessment.

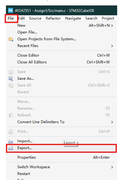
Contribution Declaration (Week 13)

It is generally expected that every student contributes his/her fair share of the workload and hence everyone in the group gets the same group component of the assignment marks. **If this is not the case** despite your best efforts to balance the workload with your partner(s) throughout the entire assignment duration, please complete the online declaration of contribution through **Canvas > [T]EE2028 > Quizzes > Assignment 2 Peer Review** after the assessment. The deadline for online declaration is **12PM (noon) 14-NOV**. The results from the online Contribution Declaration will be cross-checked with inputs from the assessing GA, and the group component for every member will then be adjusted accordingly. Your inputs will be treated with confidentiality and only accessible by the Teaching Team – your partner will not be able to see your comments or how you rated them, so you can and should be honest. The released marks also will not reflect the individual or the group components of the assignment marks.

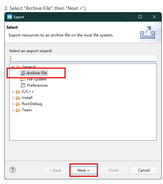
Last but not least, every team member is reminded to consciously contribute in both joint and individual efforts toward the results of this assignment and declare them clearly in the report's Appendix.

Archive project for submission

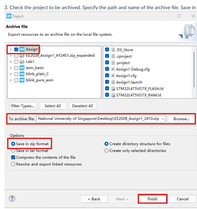
1. Go to 'File' > 'Export'.



2. Select 'Archive File', then 'Next >'.

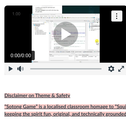


3. Check the project to be archived. Specify the path and name of the archive file. Save in zip format. Click 'Finish'.



Manually Overwrite Sensor Readings For Debug/Demo

Archive Project, Load Archived Project, and Delete Project



Disclaimer on Theme & Safety

"Sotong Game" is a localised classroom homage to "Squid Game" for educational purposes. The EE2028 teaching team does not promote or glorify violence. Thank you for keeping the spirit fun, original, and technically grounded, while showcasing your creativity.